





# **Introduction to System & Network Administrator**



# Outline

- **System administrator**
- **What does a sysadmin do?**
- **Ethical issues**
- **Administration Challenges**
- **Bugs and emergent phenomena**
- **The meta principles of system administration**
- **System Administration of Datacenter**
- **Types of Administrators/Users**

# System Administrator

- A **system administrator**, or **sysadmin**, is a **person** **who** is **responsible** for the **upkeep**, **configuration**, and **reliable** operation of **computer systems**; especially **multi-user** computers, such as **servers**.
- The system administrator seeks to **ensure** that the uptime, **performance**, **resources**, and **security** of the computers he or she **manages** meet the **needs** of the **users**, without exceeding the budget.
- To meet these needs, a system administrator may **acquire**, **install**, or **upgrade** computer **components** and **software**; provide routine **automation**; maintain security policies; troubleshoot; train or supervise staff; or offer technical support for projects.

# What does a sysadmin do?

- User account management
- Hardware management
  - Computers (Servers / Workstations)
  - Hardware (CPU, Memory, Storage, etc)
  - Network
- Software management
  - Operating System
  - Application Software (Mail service, Web service, Business software, ...)
- Perform filesystem backups, restores
- Install and configure new software and services
- Keep systems and services operating
  - Monitor system and network
  - Troubleshoot problems
- Maintain documentation
- Audit security
- Help users, performance tuning, and more!



# What does a sysadmin do?

- The subject matter of system administration includes computer **systems** and the **way people** use **them** in an **organization**. This entails a Knowledge of **Operating Systems** and **Applications**, as well as **Hardware** and **Software** troubleshooting, but also knowledge of the purposes for which people in the organization use the computers.

# User Account Management

- User **Ids**
- **Mail**
- Home **directories** (quotas, drive capacities)
- Default **startup files** (paths)
- **Permissions**, group **memberships**, **accounting** and **restrictions**
- Communicating **policies** and **procedures**
- **Disabling** / **removing** user accounts

# Hardware Management

- Capacity planning
- Inventory
- Hardware evaluation and purchase
- Adding and removing hardware
  - Configuration
  - Cabling, wiring, DIP switches, etc.
- Device driver installation
- System configuration and settings
- User notification and documentation



# Data Backups

- Perhaps most important aspect!
- Disk and backup media capacity planning
- Performance, network and system impact
- Disaster recovery
  - Onsite/Offsite
  - Periodic testing
  - Multiple copies
- User communication
  - Schedules, restore guarantees and procedures, loss tolerance

# Software Installation/Maintenance

- Evaluation of software
- Downloading and building (compiling and tweaking)
- Installation
- Maintenance of multiple versions
- Security
- Patches and updates
- User notification, documentation

# System Monitoring

- Hardware and services functioning and operational
- Capacity
  - Disk, RAM, CPU, network
- Security
  - Passwords
  - Break-ins
- System logs
  - Examination
  - Periodic rotation and truncation

# Troubleshooting

- Problem discovery, diagnosis, and resolution
  - Root cause analysis
  - Often quite difficult!
- Often requires
  - Broad and thorough system knowledge
  - Outside experts
  - Luck
- Expediency

# Local Documentation

- Administrative policies and procedures
  - Backup media locations
  - Hardware
- Location
- Description, configuration, connections
  - Software
- Install media (or download location)
- Installation, build, and configuration details
- Patches installed
- Acceptable use policies

# Security Concerns

- System logging and audit facilities
  - Evaluation and implementation
  - Monitoring and analysis
  - Traps, auditing and monitoring programs
- Unexpected or unauthorized use detection
- Monitoring of security advisories
  - Security holes and weaknesses
  - Live exploits



# Briefly

A system administrator's responsibilities might include:

1. Applying operating system updates, patches, and configuration changes.
2. Installing and configuring new hardware and software.
3. Adding, removing, or updating user account information, resetting passwords, etc.
4. System performance tuning.
  - A. Assess the problem and establish numeric values that categorize acceptable behavior.
  - B. Measure the performance of the system before modification.
  - C. Identify the part of the system that is critical for improving the performance. This is called the bottleneck.
  - D. Modify that part of the system to remove the bottleneck.
  - E. Measure the performance of the system after modification.
  - F. If the modification makes the performance better, adopt it. If the modification makes the performance worse, put it back the way it was.

# Briefly

5. Responsibility for documenting the configuration of the system.
6. Responsibility for security
7. Performing routine audits of system and software.
8. Performing backups.
9. Analyzing system logs and identifying potential issues with computer systems.
10. Troubleshooting any reported problems.
11. Introduction and integrating new technologies into existing data center environments.
12. Answering technical queries.
13. Insuring that the network infrastructure is up and running

# Ethical issues

- Because computer systems are human–computer **communities**, there are ethical considerations involved in their administration. Even if certain decisions can be made objectively, e.g. for **maximizing** productivity or **minimizing** cost, one must have a **policy** for the **use** and **management** of **computers** and **their users**. Some **decisions** have to be made to **protect** the **rights** of **individuals**.
- A system administrator has many responsibilities and constraints to consider. Ethically, ***the first responsibility must be to the greater network community, and then to the users*** them in the production of real work.

# Administration Challenges

- System administration is not just about installing operating systems. It is about planning and designing an efficient community of computers so that real users will be able to get their jobs done. That means:
  - ✓ Designing a network which is logical and efficient.
  - ✓ Deploying large numbers of machines which can be easily upgraded later.
  - ✓ Deciding what services are needed.
  - ✓ Planning and implementing adequate security.
  - ✓ Providing a comfortable environment for users.
  - ✓ Developing ways of fixing errors and problems which occur.
  - ✓ Keeping track of and understanding how to use the enormous amount of knowledge which increases every year.

# Administration Challenges

- So, system administrator's needs:
  - Broad knowledge of hardware and software
  - To balance conflicting requirements
  - Short-term vs. long-term needs
  - End-user vs. organizational requirements
  - Service provider vs. police model
    - ✓ To work well and efficiently under pressure
    - ✓ 24x7 availability
    - ✓ Flexibility, tolerance, and patience
    - ✓ Good communication skills

# Administration Challenges

*People think of sysadmins only  
when things don't work!*

# The meta principles of system administration

- The term system clearly implies an operation that is systematic, or predictable – but, unlike simple mechanical systems, like say a clock, computers interact with humans in a complex cycle of feedback, where uncertainty can enter at many levels. That makes human–computer systems difficult to predict, unless we somehow fix the boundaries of what is allowed, as a matter of policy.
- ❑ **Principle 1 (Policy is the foundation).** System administration begins with a policy – a decision about what we want and what should be, in relation to what we can afford.

# The meta principles of system administration

- ❑ **Principle 2** (**Predictability**). The highest level aim in system administration is to work towards a predictable system. Predictability has limits. It is the basis of reliability, hence trust and therefore security.
- ❑ **Principle 3** (**Scalability**). Scalable systems are those that grow in accordance with policy; i.e. they continue to function predictably, even as they increase in size.



# System Administration of Datacenter

## ▪ Duties of a Datacenter Engineer:

- Install, builds, upgrades, configuration, provisions and install servers from scratch spanning several different platforms and maintain servers as our customers grow.
- Performs basic monitoring, troubleshooting and repair all aspects of servers, operating system and hardware.
- Responds to customer impacting events. Upgrades and downgrades servers. Removes hardware from server software.
- Monitors all Electrical, Mechanical and Emergency Generator system and reports issues as appropriate as long as the Facility technician is away.
- Performs routine maintenance activities on customer server environments.
- Assists in the planning of server environments.
- Participate in analyzing and summarizing Data Center technology and critical services engineering agreements.

# System Administration of Datacenter

- Collaborate with other teams to ensure technology solutions and business needs align
- Contribute to web site design and maintenance for vendor Management team
- Support internal and external team communication plans, for instance coordinating status reporting across the team, planning team building activities, etc.
- Respond, action and resolve requests / faults logged both internally and externally, and escalation of problems and issues to the Operating Team.
- Report broken servers.
- Active monitoring of all servers.
- Work with Customer Operation Engineers in configuring customer requirements.
- Work closely with 3<sup>rd</sup> party providers.

# Types of Administrators/Users

- In a larger company, following may all be separate positions within a computer support or Information Services (IS) department. In a smaller group they may be shared by a few System Administrators, or even a single person.
  - ☐ Database Administrator
  - ☐ Network administrator
  - ☐ Security Administrator
  - ☐ Web Administrator
  - ☐ Technical support
  - ☐ Computer operator

# Database Administrator

- A **database administrator** (DBA) maintains a database system, and is responsible for the integrity of the data and the efficiency and performance of the system.



# Network Administrator

- A **network administrator** maintains network infrastructure as switches and routers, and diagnoses problems with these or with the behavior of network-attached computers.





# Security Administrator

- A **security administrators** is a specialist in computer and network security, including the administration of security devices such as firewalls, as well as consulting on general security measures.



# Web Administrator

- A **web administrator** maintains web server services (such as IIS or Apache) that allow for internal or external access to web sites. Tasks include managing multiple sites, administering security, and configuring necessary components software. Responsibility may also include software change management.



# Technical support

- **Technical support** staff respond to individual users' difficulties with computer system, provide instructions and sometimes training, and diagnose and solve common problems.





# Computer operator

- A **computer operator** performs routine maintenance and upkeep, such as changing backup tapes or replacing failed drives in a RAID array. Such tasks usually require physical presence in the room with the computer; and while less skilled than sysadmin tasks require a similar level of trust, since the operator has access to possibly sensitive data.



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# Network Administrator Tasks

- The main tasks associated with network administration include:
  - ❑ Design, installation and evaluation of the network
  - ❑ Execution and administration of regular backups
  - ❑ Creation of precise technical documentation, such as network diagrams, network cabling documents, etc.
  - ❑ Provision for precise authentication to access network resources
  - ❑ Provision for troubleshooting assistance
  - ❑ Administration of network security, including intrusion detection